

No. Of Operations

Problem 1

Steps	Operations	Iteration	Sub Total
1.	3	1	3
2.	1	N	N
3.	1	N-1	N-1
4.	2	N-2	2N-4
5.	2	N-1	2N-2
6.	2	1	2

$$= 3 + N + N-1 + 2N-4 + 2N-2 + 2$$
$$= 6N-4$$

Problem 2

Steps	Operation	Iteration	Sub Total
1.	2	1	2
2.	1	N+1	N+1
3.	2	N	2N
4.	2	N	2N
5.	1	1	1

$$= 2 + N + 1 + 2N + 2N + 1$$
$$= 4 + 5N$$

Problem 3

Steps	Operations	Iteration	Sub Total
1.	2	1	2
2.	1	$N+1$	$N+1$
3.	1	N	N
4.	2	N	$2N$
5.	2	N	$2N$
6.	1	1	1

$$= 2 + N + 1 + N + 2N + 2N + 1$$

$$= 4 + 6N$$

Problem 4

Steps	Operations	Iterations	Sub Total
1.	2	1	2
2.	1	$N+1$	$N+1$
3.	1	N	N
4.	1	N	N
5.	2	N	$2N$
6.	1	1	1

$$= 2 + N + 1 + N + N + 2N + 1$$

$$= 4 + 5N$$

Big Oh Notation

Problem 1 Solution

Analysis

'i' increases as 1, 5, 25, 125, ... (exponentially)
condition stop when $i > n$

no. of iterations = $\log_5(n) \rightarrow O(\log n)$

Answer

$O(\log n)$

Problem 2 Solution

Analysis

Outer loop (i) $\rightarrow$ runs n times

Inner loop (j) $\rightarrow$ doubles each time $\rightarrow$ values: 1, 2, 4, 8, ... $< n$
 $\rightarrow \log(n)$ times

Total = $n * \log(n)$

Answer

$O(n \log n)$

Problem 3 Solution

Analysis

Outer loop (i) $\rightarrow$ doubles each time $\rightarrow 1, 2, 4, 8, \dots \rightarrow \log_2(n)$

Inner loop (j) $\rightarrow$ triples $\rightarrow 1, 3, 9, \dots \rightarrow \log_3(n)$ times

Multiply: $\log_2(n) * \log_3(n)$

But ignore small values (logs) $\rightarrow O((\log n))^2$

Answer

$O((\log n)^2)$

Problem 4 Solution

Analysis

Outer loop (i) $\rightarrow n$ times $\rightarrow O(n)$

Middle loop (j) $\rightarrow n$ times $\rightarrow O(n)$

Inner loop (k) $\rightarrow$ doubles $\rightarrow O(\log n)$

Multiply: $n * n * \log n = n^2 \log n$

Answer

$O(n^2 \log n)$

Problem 5 Solution

Analysis

Outer while loop $\rightarrow$ halves each time $\rightarrow m = n, n/2, n/4, \dots$
 $\rightarrow$ runs $\log_2(n)$ times

Inner for loop $\rightarrow$ runs 'n' times

Total: $n * \log n$

Answer

$O(n \log n)$

Problem 6 Solution

Analysis

Outer loop (i) $\rightarrow$ goes from 1 to n

Inner loop (j) $\rightarrow$ for each 'i', runs $\log_2(i)$ times

Total Complexity: $\sum \log i$ (from $i=1$ to n)

This sum = $\log(1) + \log(2) + \dots + \log(n) = \log(n!)$

Approx: $\log(n!) = n \log n$

Answer

$O(n \log n)$

Problem 7 Solution

Analysis

Outer loop (i) $\rightarrow n$ times

Middle loop (j) $\rightarrow$ doubles $\rightarrow \log_2(n)$ times

Inner loop (k) $\rightarrow$ doubles $\rightarrow \log_2(n)$ times

Multiply : $n * \log n * \log n = n^2(\log n)^2$

$$O(n(\log n)^2)$$

Quick Reference Table

loop pattern	Complexity
for (i=1; i<=n; i++)	$O(n)$
for (i=1; i<=n; i=i*2)	$O(\log n)$
for (i=1; i<=n; i=i*5)	$O(\log n)$
for (i=n; i>1; i=i/2)	$O(\log n)$
Nested: $O(f(n)) \times O(g(n))$	$O(f(n)) \times (g(n))$

Systematic Approach

Step 1 : Identify loop pattern

$$\bullet i++ - O(n)$$

$$\bullet i = i * c - O(\log n)$$

$$\bullet i = i + c - O(n)$$

$$\bullet i = i / c - O(\log n)$$